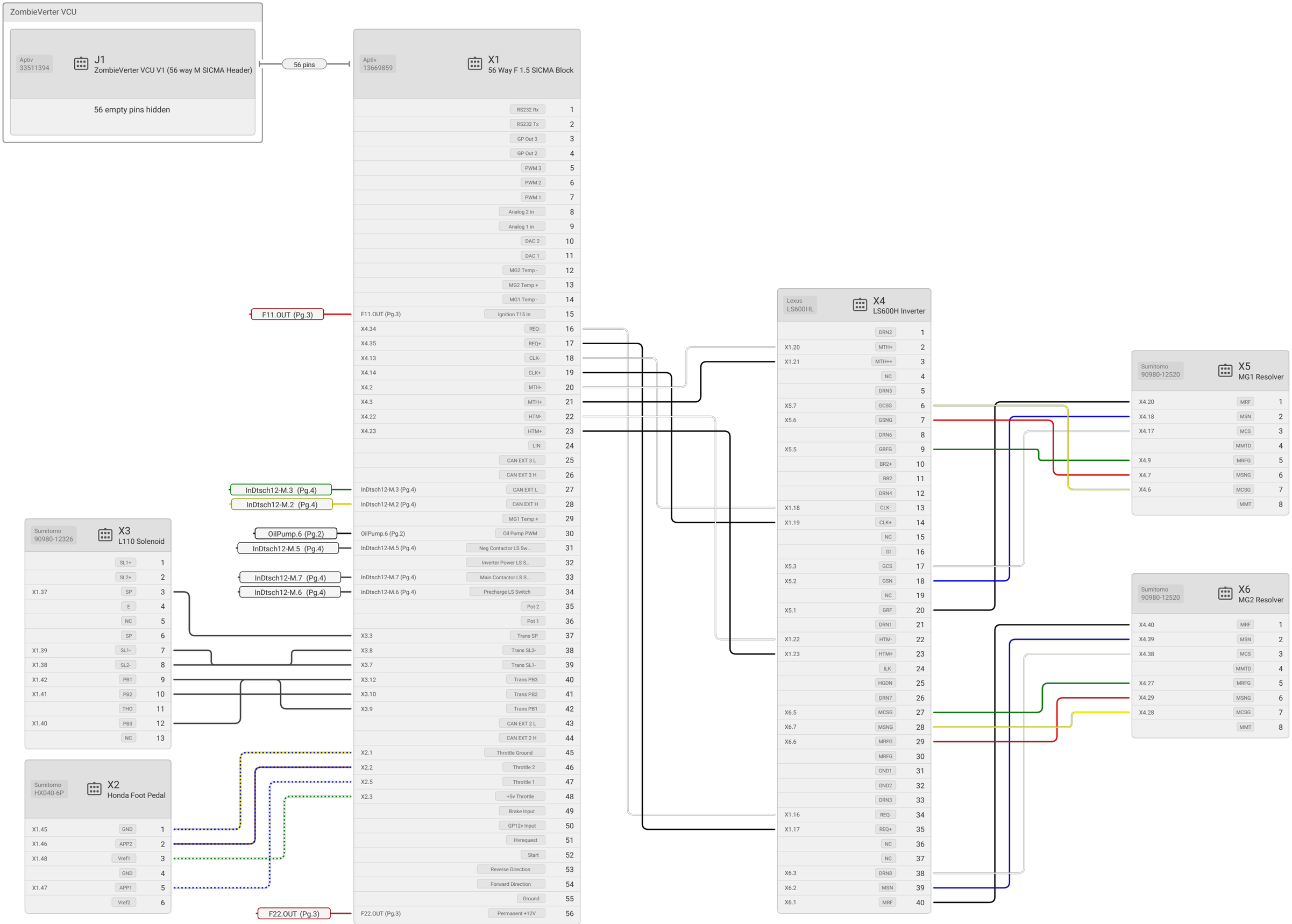
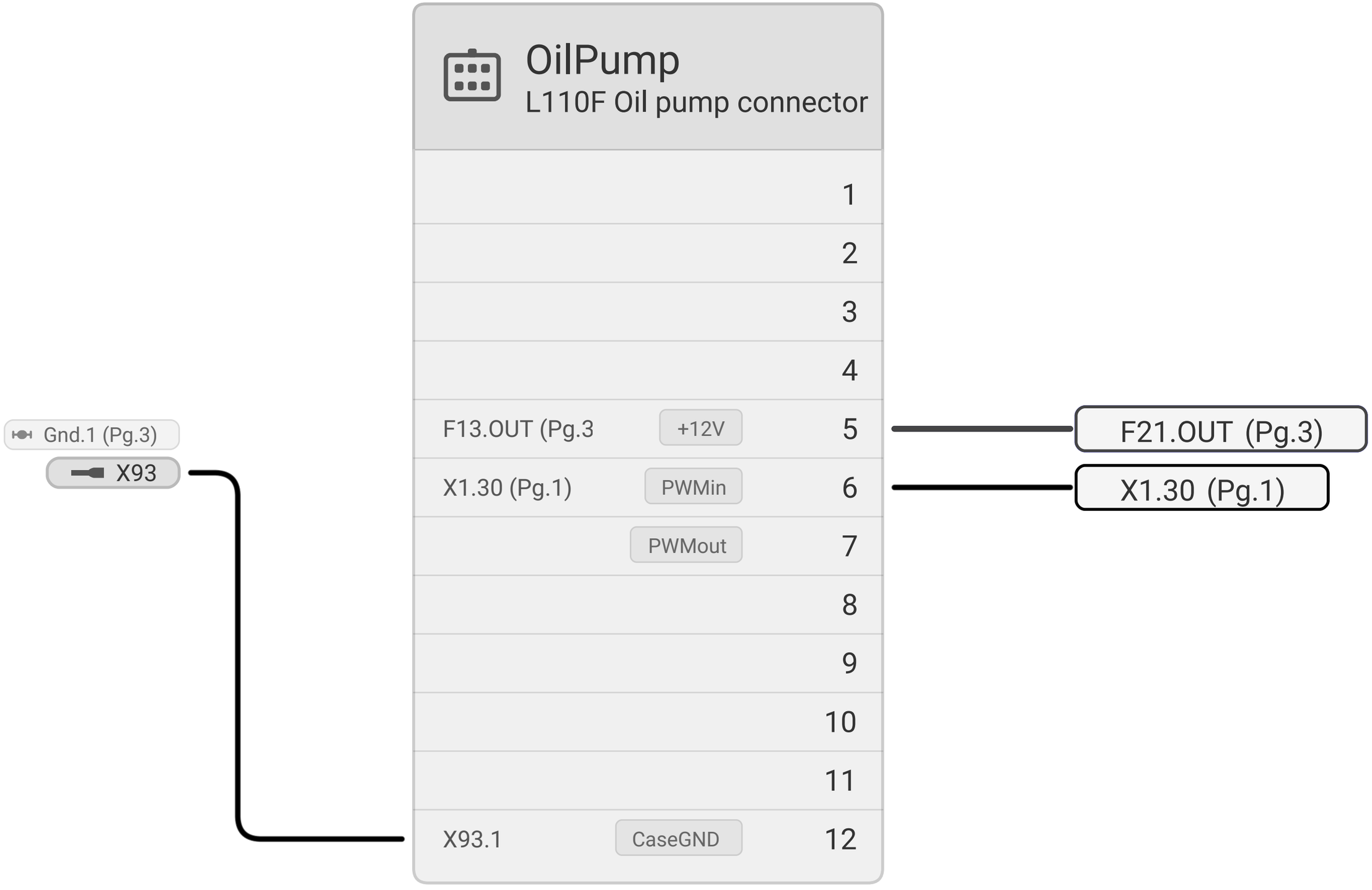
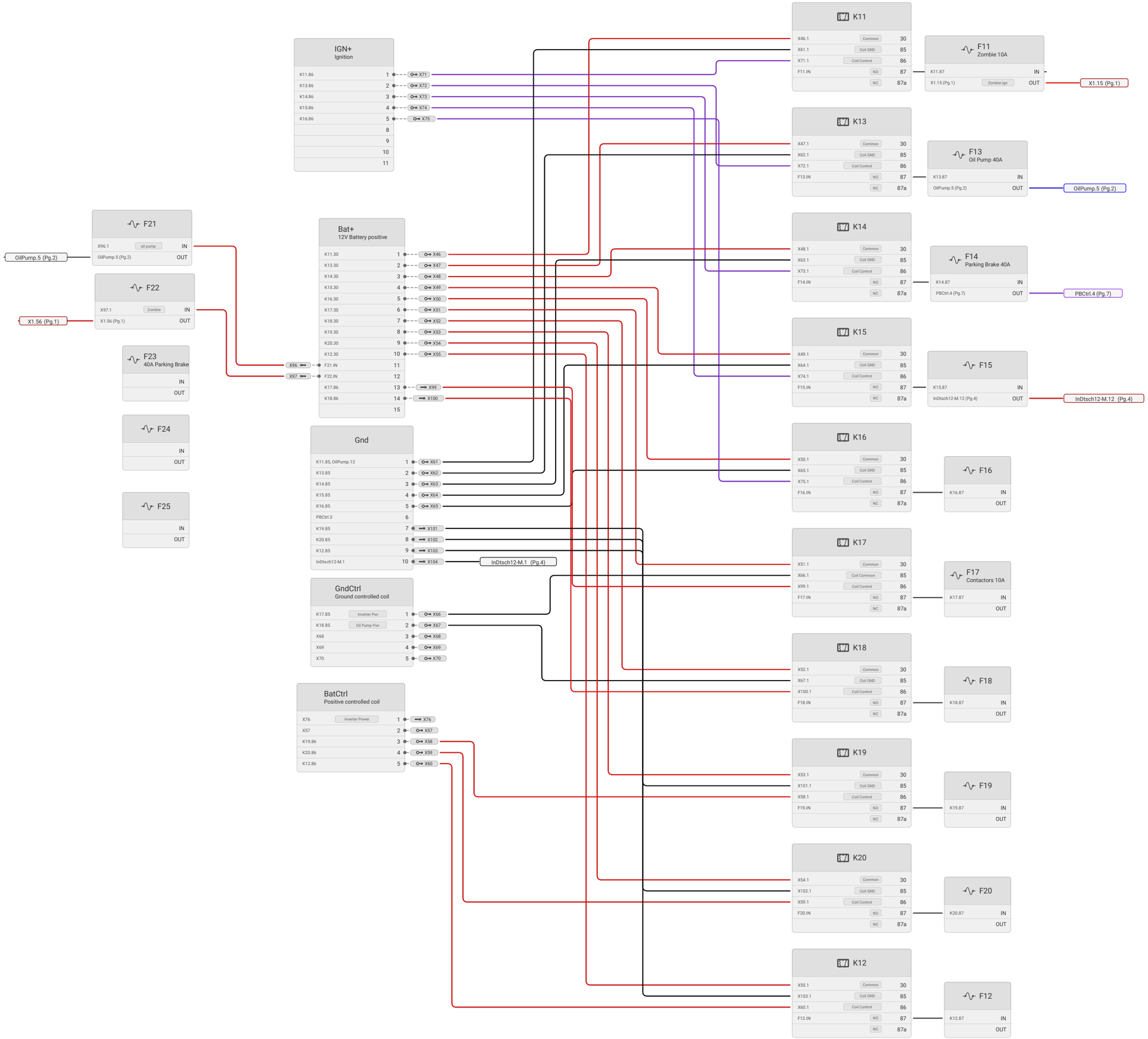
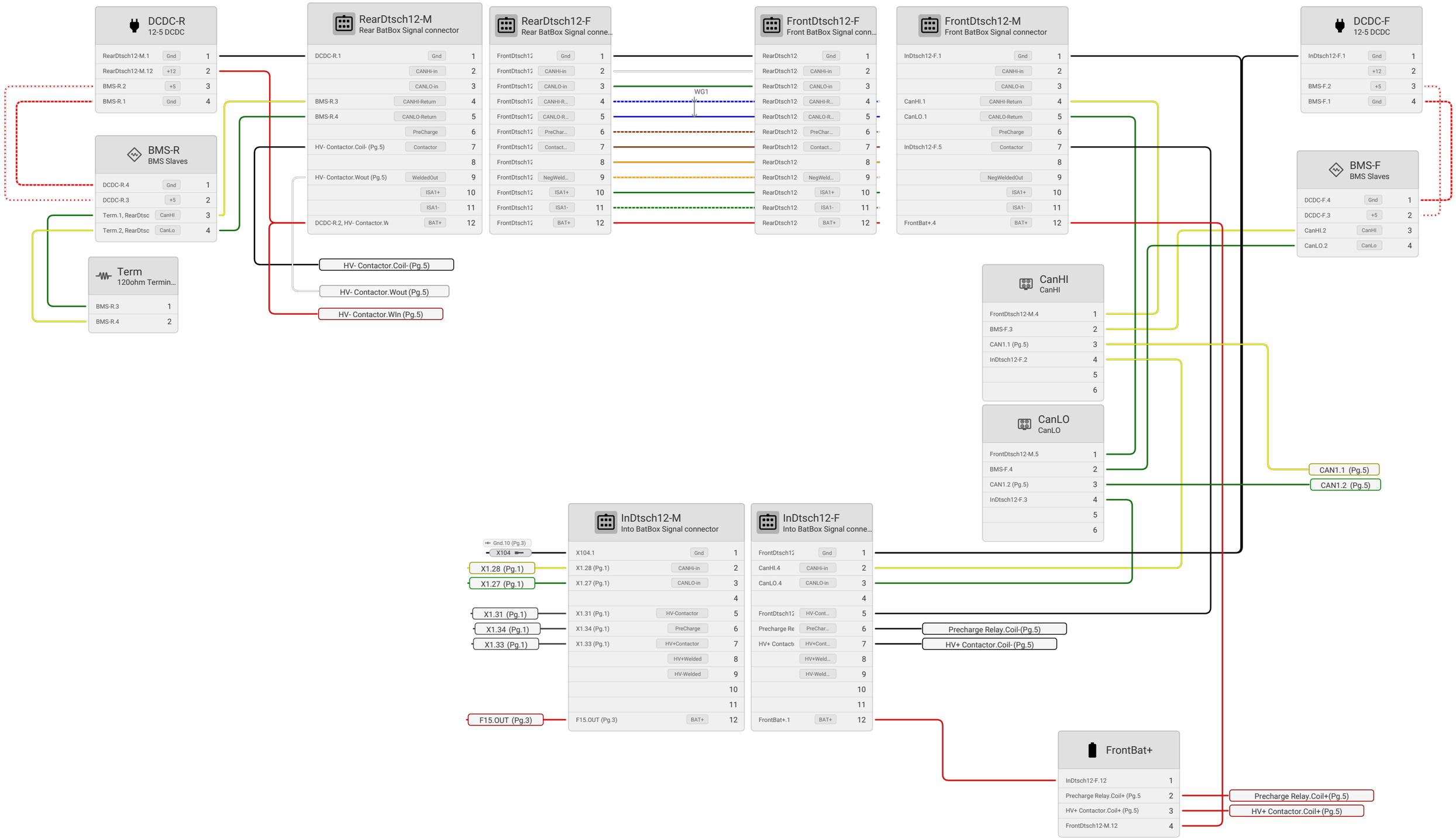
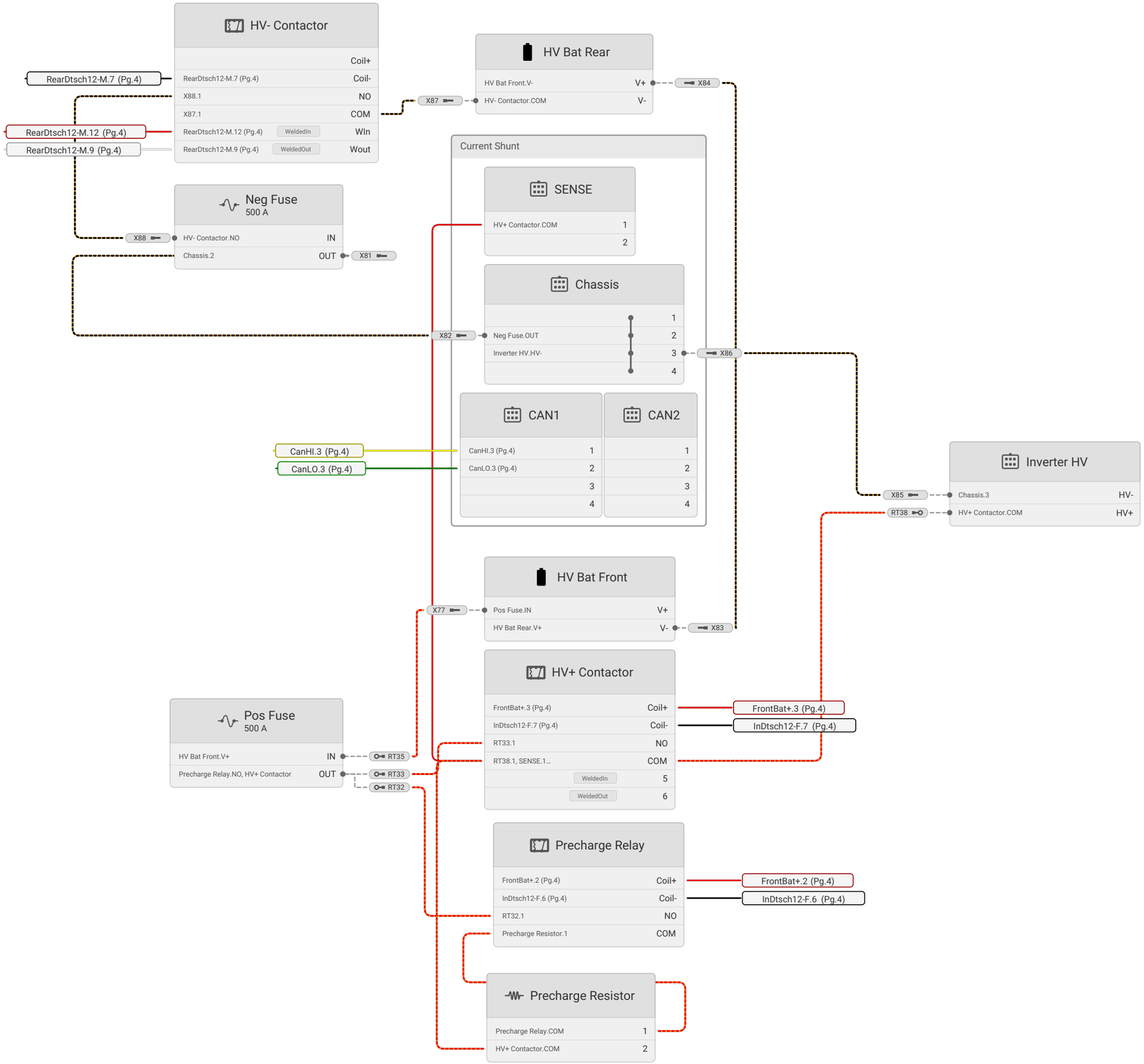
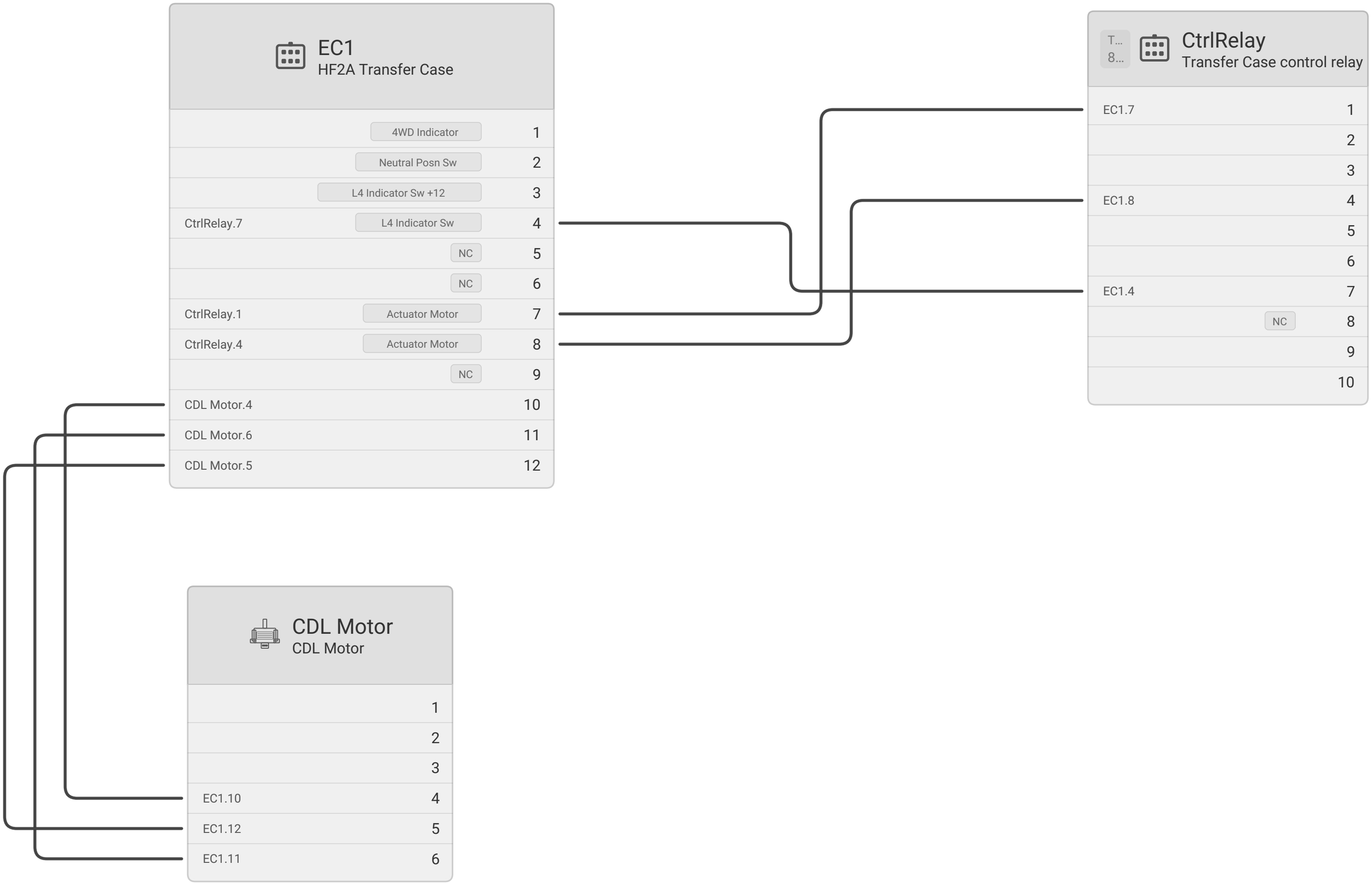

ZombieVerter VCU
Aptiv 33511394
J1
ZombieVerter VCU V1 (56 way M SICMA Header)
56 empty pins hidden
56 pins
Aptiv 13669859
X1
56 Way F 1.5 SICMA Block
RS232 Rx 1
RS232 Tx 2
GP Out 3 3
GP Out 2 4
PWM 3 5
PWM 2 6
PWM 1 7
Analog 2 in 8
Analog 1 in 9
DAC 2 10
DAC 1 11
MG2 Temp - 12
MG2 Temp + 13
MG1 Temp - 14
F11.OUT (Pg.3) Ignition T15 In 15
X4.34 REQ- 16
X4.35 REQ+ 17
X4.13 CLK- 18
X4.14 CLK+ 19
X4.2 MTH- 20
X4.3 MTH+ 21
X4.22 HTM- 22
X4.23 HTM+ 23
LIN 24
CAN EXT 3 L 25
CAN EXT 3 H 26
InDtsch12-M.3 (Pg.4) CAN EXT L 27
InDtsch12-M.2 (Pg.4) CAN EXT H 28
MG1 Temp + 29
OilPump.6 (Pg.2) Oil Pump PWM 30
InDtsch12-M.5 (Pg.4) Neg Contactor LS Sw... 31
Inverter Power LS S... 32
InDtsch12-M.7 (Pg.4) Main Contactor LS S... 33
InDtsch12-M.6 (Pg.4) Precharge LS Switch 34
Pot 2 35
Pot 1 36
X3.3 Trans SP 37
X3.8 Trans SL2- 38
X3.7 Trans SL1- 39
X3.12 Trans PB3 40
X3.10 Trans PB2 41
X3.9 Trans PB1 42
CAN EXT 2 L 43
CAN EXT 2 H 44
X2.1 Throttle Ground 45
X2.2 Throttle 2 46
X2.5 Throttle 1 47
X2.3 +5v Throttle 48
Brake Input 49
GP12v Input 50
Hvrequest 51
Start 52
Reverse Direction 53
Forward Direction 54
Ground 55
F22.OUT (Pg.3) Permanent +12V 56
F11.OUT (Pg.3)
InDtsch12-M.3 (Pg.4)
InDtsch12-M.2 (Pg.4)
OilPump.6 (Pg.2)
InDtsch12-M.5 (Pg.4)
InDtsch12-M.7 (Pg.4)
InDtsch12-M.6 (Pg.4)
F22.OUT (Pg.3)
Sumitomo 90980-12326
X3
L110 Solenoid
SL1+ 1
SL2+ 2
X1.37 SP 3
E 4
NC 5
SP 6
X1.39 SL1- 7
X1.38 SL2- 8
X1.42 PB1 9
X1.41 PB2 10
THO 11
X1.40 PB3 12
NC 13
Sumitomo HX040-6P
X2
Honda Foot Pedal
X1.45 GND 1
X1.46 APP2 2
X1.48 Vref1 3
GND 4
X1.47 APP1 5
Vref2 6
Lexus LS600HL
X4
LS600H Inverter
DRN2 1
X1.20 MTH+ 2
X1.21 MTH++ 3
NC 4
DRN5 5
X5.7 GCSG 6
X5.6 GSNG 7
DRN6 8
X5.5 GRFG 9
BR2+ 10
BR2 11
DRN4 12
X1.18 CLK- 13
X1.19 CLK+ 14
NC 15
GI 16
X5.3 GCS 17
X5.2 GSN 18
NC 19
X5.1 GRF 20
DRN1 21
X1.22 HTM- 22
X1.23 HTM+ 23
ILK 24
HGDN 25
DRN7 26
X6.5 MCSG 27
X6.7 MSNG 28
X6.6 MRFG 29
MRFG 30
GND1 31
GND2 32
DRN3 33
X1.16 REQ- 34
X1.17 REQ+ 35
NC 36
NC 37
X6.3 DRN8 38
X6.2 MSN 39
X6.1 MRF 40
Sumitomo 90980-12520
X5
MG1 Resolver
X4.20 MRF 1
X4.18 MSN 2
X4.17 MCS 3
MMTD 4
X4.9 MRFG 5
X4.7 MSNG 6
X4.6 MCSG 7
MMT 8
Sumitomo 90980-12520
X6
MG2 Resolver
X4.40 MRF 1
X4.39 MSN 2
X4.38 MCS 3
MMTD 4
X4.27 MRFG 5
X4.29 MSNG 6
X4.28 MCSG 7
MMT 8

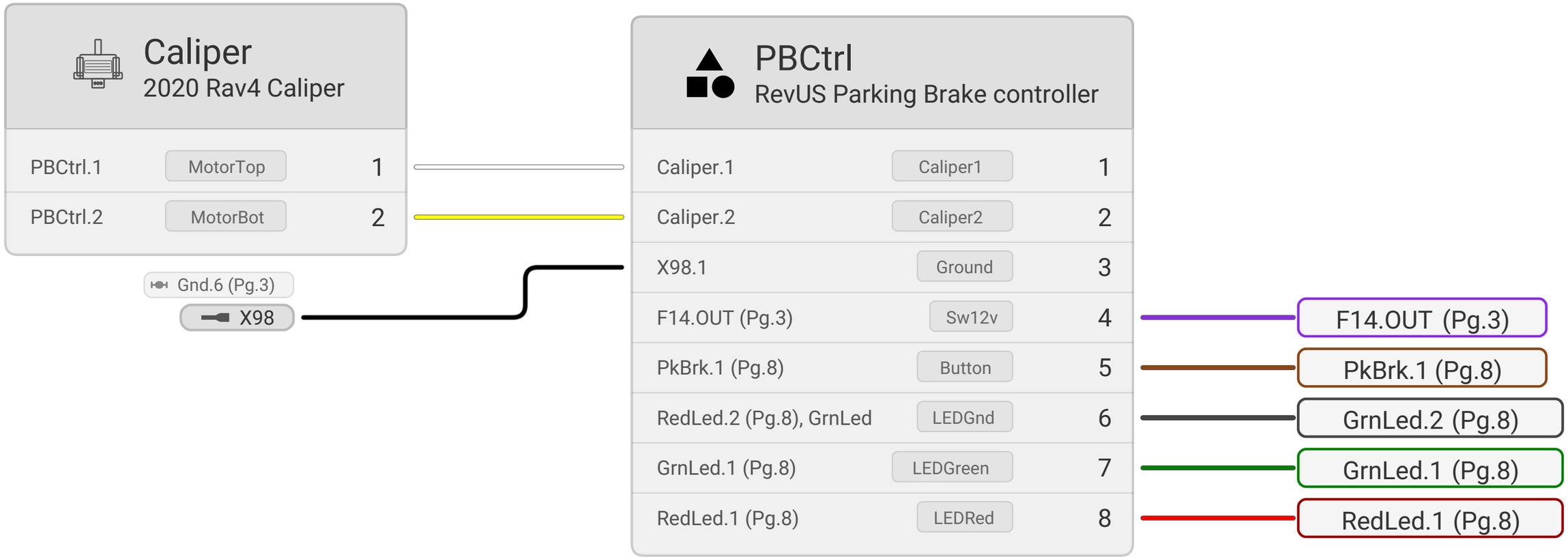

OilPump
L110F Oil pump connector
1
2
3
4
F13.OUT (Pg.3
+12V
5
F21.OUT (Pg.3)
X1.30 (Pg.1)
PWMin
6
X1.30 (Pg.1)
PWMout
7
8
9
10
11
X93.1
CaseGND
12
Gnd.1 (Pg.3)
X93

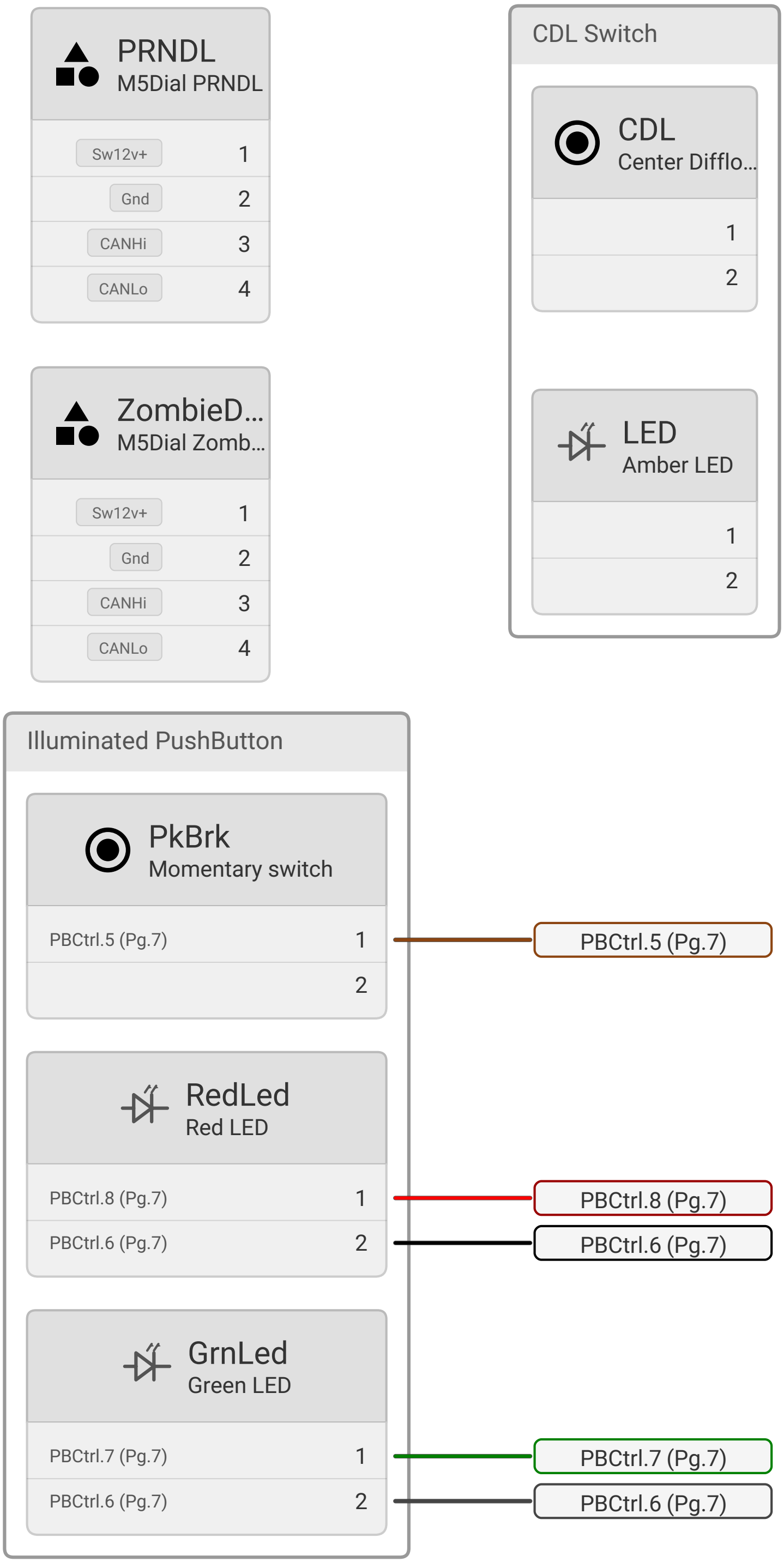

IGN+
Ignition
K11.86 1 X71
K13.86 2 X72
K14.86 3 X73
K15.86 4 X74
K16.86 5 X75
8
9
10
11
Bat+
12V Battery positive
K11.30 1 X46
K13.30 2 X47
K14.30 3 X48
K15.30 4 X49
K16.30 5 X50
K17.30 6 X51
K18.30 7 X52
K19.30 8 X53
K20.30 9 X54
K12.30 10 X55
F21.IN 11
F22.IN 12
K17.86 13 X99
K18.86 14 X100
15
Gnd
K11.85, OilPump.12 1 X61
K13.85 2 X62
K14.85 3 X63
K15.85 4 X64
K16.85 5 X65
PBCtrl.3 6
K19.85 7 X101
K20.85 8 X102
K12.85 9 X103
InDtsch12-M.1 10 X104
InDtsch12-M.1 (Pg.4)
GndCtrl
Ground controlled coil
K17.85 Inverter Pwr 1 X66
K18.85 Oil Pump Pwr 2 X67
X68 3 X68
X69 4 X69
X70 5 X70
BatCtrl
Positive controlled coil
X76 Inverter Power 1 X76
X57 2 X57
K19.86 3 X58
K20.86 4 X59
K12.86 5 X60
F21
X96.1 oil pump IN
OilPump.5 (Pg.2) OUT
OilPump.5 (Pg.2)
F22
X97.1 Zombie IN
X1.56 (Pg.1) OUT
X1.56 (Pg.1)
F23
40A Parking Brake
IN
OUT
F24
IN
OUT
F25
IN
OUT
X96
X97
K11
X46.1 Common 30
X61.1 Coil GND 85
X71.1 Coil Control 86
F11.IN NO 87
NC 87a
F11
Zombie 10A
K11.87 IN
X1.15 (Pg.1) Zombie Ign OUT
X1.15 (Pg.1)
K13
X47.1 Common 30
X62.1 Coil GND 85
X72.1 Coil Control 86
F13.IN NO 87
NC 87a
F13
Oil Pump 40A
K13.87 IN
OilPump.5 (Pg.2) OUT
OilPump.5 (Pg.2)
K14
X48.1 Common 30
X63.1 Coil GND 85
X73.1 Coil Control 86
F14.IN NO 87
NC 87a
F14
Parking Brake 40A
K14.87 IN
PBCtrl.4 (Pg.7) OUT
PBCtrl.4 (Pg.7)
K15
X49.1 Common 30
X64.1 Coil GND 85
X74.1 Coil Control 86
F15.IN NO 87
NC 87a
F15
K15.87 IN
InDtsch12-M.12 (Pg.4) OUT
InDtsch12-M.12 (Pg.4)
K16
X50.1 Common 30
X65.1 Coil GND 85
X75.1 Coil Control 86
F16.IN NO 87
NC 87a
F16
K16.87 IN
OUT
K17
X51.1 Common 30
X66.1 Coil Common 85
X99.1 Coil Control 86
F17.IN NO 87
NC 87a
F17
Contactors 10A
K17.87 IN
OUT
K18
X52.1 Common 30
X67.1 Coil GND 85
X100.1 Coil Control 86
F18.IN NO 87
NC 87a
F18
K18.87 IN
OUT
K19
X53.1 Common 30
X101.1 Coil GND 85
X58.1 Coil Control 86
F19.IN NO 87
NC 87a
F19
K19.87 IN
OUT
K20
X54.1 Common 30
X102.1 Coil GND 85
X59.1 Coil Control 86
F20.IN NO 87
NC 87a
F20
K20.87 IN
OUT
K12
X55.1 Common 30
X103.1 Coil GND 85
X60.1 Coil Control 86
F12.IN NO 87
NC 87a
F12
K12.87 IN
OUT


DCDC-R
12-5 DCDC
RearDtsch12-M.1 Gnd 1
RearDtsch12-M.12 +12 2
BMS-R.2 +5 3
BMS-R.1 Gnd 4
BMS-R
BMS Slaves
DCDC-R.4 Gnd 1
DCDC-R.3 +5 2
Term.1, RearDtsc CanHI 3
Term.2, RearDtsc CanLo 4
Term
120ohm Termin...
BMS-R.3 1
BMS-R.4 2
RearDtsch12-M
Rear BatBox Signal connector
DCDC-R.1 Gnd 1
CANHi-in 2
CANLO-in 3
BMS-R.3 CANHI-Return 4
BMS-R.4 CANLO-Return 5
PreCharge 6
HV- Contactor.Coil- (Pg.5) Contactor 7
8
HV- Contactor.Wout (Pg.5) WeldedOut 9
ISA1+ 10
ISA1- 11
DCDC-R.2, HV- Contactor.W BAT+ 12
RearDtsch12-F
Rear BatBox Signal conne...
FrontDtsch12 Gnd 1
FrontDtsch12 CANHi-in 2
FrontDtsch12 CANLO-in 3
FrontDtsch12 CANHI-R... 4
FrontDtsch12 CANLO-R... 5
FrontDtsch12 PreChar... 6
FrontDtsch12 Contact... 7
FrontDtsch12 8
FrontDtsch12 NegWeld... 9
FrontDtsch12 ISA1+ 10
FrontDtsch12 ISA1- 11
FrontDtsch12 BAT+ 12
WG1
FrontDtsch12-F
Front BatBox Signal conn...
RearDtsch12 Gnd 1
RearDtsch12 CANHi-in 2
RearDtsch12 CANLO-in 3
RearDtsch12 CANHI-R... 4
RearDtsch12 CANLO-R... 5
RearDtsch12 PreChar... 6
RearDtsch12 Contact... 7
RearDtsch12 8
RearDtsch12 NegWeld... 9
RearDtsch12 ISA1+ 10
RearDtsch12 ISA1- 11
RearDtsch12 BAT+ 12
FrontDtsch12-M
Front BatBox Signal connector
InDtsch12-F.1 Gnd 1
CANHi-in 2
CANLO-in 3
CanHI.1 CANHI-Return 4
CanLO.1 CANLO-Return 5
PreCharge 6
InDtsch12-F.5 Contactor 7
8
NegWeldedOut 9
ISA1+ 10
ISA1- 11
FrontBat+.4 BAT+ 12
DCDC-F
12-5 DCDC
InDtsch12-F.1 Gnd 1
+12 2
BMS-F.2 +5 3
BMS-F.1 Gnd 4
BMS-F
BMS Slaves
DCDC-F.4 Gnd 1
DCDC-F.3 +5 2
CanHI.2 CanHI 3
CanLO.2 CanLo 4
CanHI
CanHI
FrontDtsch12-M.4 1
BMS-F.3 2
CAN1.1 (Pg.5) 3
InDtsch12-F.2 4
5
6
CanLO
CanLO
FrontDtsch12-M.5 1
BMS-F.4 2
CAN1.2 (Pg.5) 3
InDtsch12-F.3 4
5
6
CAN1.1 (Pg.5)
CAN1.2 (Pg.5)
HV- Contactor.Coil- (Pg.5)
HV- Contactor.Wout (Pg.5)
HV- Contactor.WIn (Pg.5)
InDtsch12-M
Into BatBox Signal connector
Gnd.10 (Pg.3)
X104
X104.1 Gnd 1
X1.28 (Pg.1)
X1.28 (Pg.1) CANHi-in 2
X1.27 (Pg.1)
X1.27 (Pg.1) CANLO-in 3
4
X1.31 (Pg.1)
X1.31 (Pg.1) HV-Contactor 5
X1.34 (Pg.1)
X1.34 (Pg.1) PreCharge 6
X1.33 (Pg.1)
X1.33 (Pg.1) HV+Contactor 7
HV+Welded 8
HV-Welded 9
10
11
F15.OUT (Pg.3)
F15.OUT (Pg.3) BAT+ 12
InDtsch12-F
Into BatBox Signal conne...
FrontDtsch12 Gnd 1
CanHI.4 CANHi-in 2
CanLO.4 CANLO-in 3
4
FrontDtsch12 HV-Cont... 5
Precharge Re PreChar... 6
HV+ Contacto HV+Cont... 7
HV+Weld... 8
HV-Weld... 9
10
11
FrontBat+.1 BAT+ 12
Precharge Relay.Coil-(Pg.5)
HV+ Contactor.Coil- (Pg.5)
FrontBat+
InDtsch12-F.12 1
Precharge Relay.Coil+ (Pg.5 2
HV+ Contactor.Coil+ (Pg.5) 3
FrontDtsch12-M.12 4
Precharge Relay.Coil+(Pg.5)
HV+ Contactor.Coil+ (Pg.5)


HV- Contactor
Coil+
RearDtsch12-M.7 (Pg.4)
RearDtsch12-M.7 (Pg.4)
Coil-
X88.1
NO
X87.1
COM
RearDtsch12-M.12 (Pg.4)
RearDtsch12-M.12 (Pg.4)
WeldedIn
WIn
RearDtsch12-M.9 (Pg.4)
RearDtsch12-M.9 (Pg.4)
WeldedOut
Wout
HV Bat Rear
HV Bat Front.V-
V+
X84
X87
HV- Contactor.COM
V-
Current Shunt
SENSE
HV+ Contactor.COM
1
2
Neg Fuse
500 A
X88
HV- Contactor.NO
IN
Chassis.2
OUT
X81
Chassis
1
X82
Neg Fuse.OUT
2
Inverter HV.HV-
3
X86
4
CAN1
CAN2
CanHI.3 (Pg.4)
CanLO.3 (Pg.4)
CanHI.3 (Pg.4)
1
1
CanLO.3 (Pg.4)
2
2
3
3
4
4
Inverter HV
X85
Chassis.3
HV-
RT38
HV+ Contactor.COM
HV+
HV Bat Front
X77
Pos Fuse.IN
V+
HV Bat Rear.V+
V-
X83
HV+ Contactor
FrontBat+.3 (Pg.4)
Coil+
FrontBat+.3 (Pg.4)
InDtsch12-F.7 (Pg.4)
Coil-
InDtsch12-F.7 (Pg.4)
RT33.1
NO
RT38.1, SENSE.1...
COM
WeldedIn
5
WeldedOut
6
Pos Fuse
500 A
HV Bat Front.V+
IN
RT35
Precharge Relay.NO, HV+ Contactor
OUT
RT33
RT32
Precharge Relay
FrontBat+.2 (Pg.4)
Coil+
FrontBat+.2 (Pg.4)
InDtsch12-F.6 (Pg.4)
Coil-
InDtsch12-F.6 (Pg.4)
RT32.1
NO
Precharge Resistor.1
COM
Precharge Resistor
Precharge Relay.COM
1
HV+ Contactor.COM
2


EC1
HF2A Transfer Case
4WD Indicator 1
Neutral Posn Sw 2
L4 Indicator Sw +12 3
CtrlRelay.7 L4 Indicator Sw 4
NC 5
NC 6
CtrlRelay.1 Actuator Motor 7
CtrlRelay.4 Actuator Motor 8
NC 9
CDL Motor.4 10
CDL Motor.6 11
CDL Motor.5 12
T...
8...
CtrlRelay
Transfer Case control relay
EC1.7 1
2
3
EC1.8 4
5
6
EC1.4 7
NC 8
9
10
CDL Motor
CDL Motor
1
2
3
EC1.10 4
EC1.12 5
EC1.11 6


Caliper
2020 Rav4 Caliper
PBCtrl.1
MotorTop
1
PBCtrl.2
MotorBot
2
Gnd.6 (Pg.3)
X98
PBCtrl
RevUS Parking Brake controller
Caliper.1
Caliper1
1
Caliper.2
Caliper2
2
X98.1
Ground
3
F14.OUT (Pg.3)
Sw12v
4
PkBrk.1 (Pg.8)
Button
5
RedLed.2 (Pg.8), GrnLed
LEDGnd
6
GrnLed.1 (Pg.8)
LEDGreen
7
RedLed.1 (Pg.8)
LEDRed
8
F14.OUT (Pg.3)
PkBrk.1 (Pg.8)
GrnLed.2 (Pg.8)
GrnLed.1 (Pg.8)
RedLed.1 (Pg.8)


PRNDL
M5Dial PRNDL
Sw12v+ 1
Gnd 2
CANHi 3
CANLo 4
ZombieD...
M5Dial Zomb...
Sw12v+ 1
Gnd 2
CANHi 3
CANLo 4
CDL Switch
CDL
Center Difflo...
1
2
LED
Amber LED
1
2
Illuminated PushButton
PkBrk
Momentary switch
PBCtrl.5 (Pg.7) 1
2
PBCtrl.5 (Pg.7)
RedLed
Red LED
PBCtrl.8 (Pg.7) 1
PBCtrl.6 (Pg.7) 2
PBCtrl.8 (Pg.7)
PBCtrl.6 (Pg.7)
GrnLed
Green LED
PBCtrl.7 (Pg.7) 1
PBCtrl.6 (Pg.7) 2
PBCtrl.7 (Pg.7)
PBCtrl.6 (Pg.7)